

---

# TN3: QUANTUM-POINTER HAMILTONIAN REALISATION OF THE BELL-SINGLET CORRELATION

---

A PREPRINT

**Zayn Blore**

 zayn.blore@ou.ac.uk   
[www.constraintsurfacedynamics.com](http://www.constraintsurfacedynamics.com)

May 16, 2026

## ABSTRACT

Technical Note 2 established, non-constructively, that the two-qubit singlet lies within the finite-dimensional Born-weight layer and that the associated operational probabilities give

$$E(a, b) = -a \cdot b.$$

The present note exhibits the standard von Neumann measurement Hamiltonian as the explicit quantum-pointer realisation of that result within the finite-dimensional geometric framework. Its contribution is compatibility and architectural closure: the canonical quantum measurement construction translates directly into the system–apparatus setting without invoking Fubini–Study Voronoi volumes, isotropy-to-linearity arguments, or an alternative measurement formalism.

We consider a two-qubit system coupled to two pointer sectors. In the finite-dimensional version used here, the pointer spaces are large truncations of continuous position-pointer Hilbert spaces, so the joint system–apparatus state space is the compact projective manifold

$$\mathbb{CP}^{N-1}, \quad N = 4d_A d_B.$$

For detector settings  $a, b \in S^2$ , the impulsive readout Hamiltonian is

$$\hat{H}_{a,b} = \kappa_A(\sigma \cdot a \otimes \mathbb{I}) \otimes \hat{p}_A + \kappa_B(\mathbb{I} \otimes \sigma \cdot b) \otimes \hat{p}_B,$$

with free pointer evolution neglected during the readout interval. Through the Ashtekar–Schilling correspondence, this Schrödinger evolution is equivalently Hamiltonian flow on the joint projective manifold.

Starting from

$$|\Psi_0\rangle = |\psi^-\rangle \otimes |\phi_A^0\rangle \otimes |\phi_B^0\rangle,$$

the interaction separates the four pointer branches associated with outcomes  $s, t \in \{\pm 1\}$ . Pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \widehat{Q}_A^s \otimes \widehat{Q}_B^t$$

define the outcome sectors for the measurement context. In the strong-readout limit, their branch weights are

$$P_{st}(a, b) = \langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle = \frac{1 - st a \cdot b}{4}.$$

Hence

$$E(a, b) = \sum_{s,t=\pm 1} st P_{st}(a, b) = -a \cdot b.$$

The local marginals satisfy

$$P_A(\pm 1 \mid a, b) = P_B(\pm 1 \mid a, b) = \frac{1}{2},$$

so operational no-signalling follows by direct marginal summation.

In the CSD interpretive layer, the construction is deterministic per realised measurement context but not Bell-local: it uses context-indexed outcome sectors rather than factorised outcome functions  $R_A(\lambda, a)$  and  $R_B(\lambda, b)$ . The result should therefore be read as a clean embedding of the standard quantum-pointer singlet measurement into the finite-dimensional geometric reconstruction, and as the constructive closure of the singlet specialisation downstream of the non-constructive probability calculation.

## Contents

|          |                                                               |           |
|----------|---------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                           | <b>4</b>  |
| 1.1      | Purpose and Scope . . . . .                                   | 4         |
| 1.2      | Relation to Technical Note 2 . . . . .                        | 4         |
| 1.3      | Contribution of the Present Note . . . . .                    | 5         |
| 1.4      | Structure of the Paper . . . . .                              | 6         |
| <b>2</b> | <b>Joint System-Apparatus Geometry</b>                        | <b>6</b>  |
| 2.1      | Two-Qubit System . . . . .                                    | 6         |
| 2.2      | Finite-Dimensional Pointer Sectors . . . . .                  | 7         |
| 2.3      | Joint Projective State Space . . . . .                        | 7         |
| 2.4      | Hamiltonian Flow in the Geometric Formulation . . . . .       | 8         |
| 2.5      | Finite-Truncation Scope . . . . .                             | 8         |
| <b>3</b> | <b>Von Neumann Measurement Hamiltonian</b>                    | <b>9</b>  |
| 3.1      | Detector Settings and Spin Operators . . . . .                | 9         |
| 3.2      | Impulsive Readout Hamiltonian . . . . .                       | 9         |
| 3.3      | Commutation and Unitary Factorisation . . . . .               | 10        |
| 3.4      | Parameter Locality at the Hamiltonian-Control Level . . . . . | 10        |
| 3.5      | Free Evolution and the Impulsive Limit . . . . .              | 11        |
| <b>4</b> | <b>Branch Separation Under Hamiltonian Flow</b>               | <b>11</b> |
| 4.1      | Initial Singlet-Pointer State . . . . .                       | 11        |
| 4.2      | Expansion in the Joint Measurement Basis . . . . .            | 11        |
| 4.3      | Pointer Translation by the Measurement Interaction . . . . .  | 12        |
| 4.4      | Branch-Separated Final State . . . . .                        | 12        |
| 4.5      | Asymptotic Branch-Separation Lemma . . . . .                  | 13        |
| 4.6      | Finite-Separation Error Terms . . . . .                       | 14        |
| <b>5</b> | <b>Pointer-Induced Outcome Sectors</b>                        | <b>14</b> |
| 5.1      | Pointer Readout Projectors . . . . .                          | 14        |
| 5.2      | Outcome Sectors for a Measurement Context . . . . .           | 15        |
| 5.3      | Operator-Form Branch Weights . . . . .                        | 15        |
| 5.4      | CSD Basin Interpretation . . . . .                            | 16        |
| 5.5      | No Use of Fubini-Study Voronoi Volumes . . . . .              | 16        |

|           |                                                                    |           |
|-----------|--------------------------------------------------------------------|-----------|
| 5.6       | Weak Operational Consistency for Shared Pointer Channels . . . . . | 17        |
| <b>6</b>  | <b>Singlet Probabilities and Correlation</b>                       | <b>17</b> |
| 6.1       | Projector Probabilities from the Singlet State . . . . .           | 17        |
| 6.2       | Relation Between Amplitudes and Projector Expectations . . . . .   | 18        |
| 6.3       | Evaluation of $P_{st}(a, b)$ . . . . .                             | 19        |
| 6.4       | Derivation of $E(a, b) = -a \cdot b$ . . . . .                     | 19        |
| 6.5       | Finite-Separation Corrections . . . . .                            | 20        |
| <b>7</b>  | <b>Operational No-Signalling</b>                                   | <b>20</b> |
| 7.1       | Marginal Probabilities at $A$ . . . . .                            | 20        |
| 7.2       | Marginal Probabilities at $B$ . . . . .                            | 21        |
| 7.3       | No-Signalling in the Strong-Readout Limit . . . . .                | 21        |
| 7.4       | Finite-Separation Corrections to Marginals . . . . .               | 21        |
| 7.5       | Interpretation . . . . .                                           | 22        |
| <b>8</b>  | <b>Bell Consistency</b>                                            | <b>22</b> |
| 8.1       | Context-Indexed Outcome Sectors . . . . .                          | 22        |
| 8.2       | Absence of Bell-Local Factorisation . . . . .                      | 23        |
| 8.3       | No Global Joint Assignment . . . . .                               | 23        |
| 8.4       | CHSH Value in the Pointer-Sector Construction . . . . .            | 23        |
| 8.5       | Relation to the Fine-Bell Analysis . . . . .                       | 24        |
| <b>9</b>  | <b>Discussion</b>                                                  | <b>24</b> |
| 9.1       | What Has Been Established . . . . .                                | 24        |
| 9.2       | Compatibility Rather Than Novel Measurement Theory . . . . .       | 25        |
| 9.3       | Relation to Bohmian Branch Selection . . . . .                     | 26        |
| 9.4       | Limitations and Non-Claims . . . . .                               | 26        |
| 9.5       | Role Within the Finite-Dimensional Reconstruction . . . . .        | 26        |
| <b>10</b> | <b>Conclusion</b>                                                  | <b>27</b> |
| <b>A</b>  | <b>Finite-Dimensional Pointer Truncation</b>                       | <b>28</b> |
| A.1       | Truncated Pointer Hilbert Spaces . . . . .                         | 28        |
| A.2       | Approximation to Continuous Position Pointers . . . . .            | 29        |
| A.3       | Compactness of the Joint Projective Manifold . . . . .             | 30        |
| <b>B</b>  | <b>Finite-Separation Error Bounds</b>                              | <b>30</b> |
| B.1       | Pointer Leakage Parameters . . . . .                               | 30        |
| B.2       | Error Bound for Branch Weights . . . . .                           | 30        |
| B.3       | Error Bound for the Correlator . . . . .                           | 32        |
| B.4       | Error Bound for Marginals . . . . .                                | 32        |

|          |                                             |           |
|----------|---------------------------------------------|-----------|
| B.5      | Interpretation of the Error Terms . . . . . | 33        |
| <b>C</b> | <b>Relation to the CSD Dependency Stack</b> | <b>33</b> |
| C.1      | Relation to Technical Note 2 . . . . .      | 33        |
| C.2      | Relation to Paper B . . . . .               | 34        |
| C.3      | Relation to TN4 . . . . .                   | 34        |
| C.4      | Relation to c1 . . . . .                    | 35        |
| C.5      | Relation to LF3 . . . . .                   | 35        |
| C.6      | Dependency Summary . . . . .                | 36        |

## 1 Introduction

### 1.1 Purpose and Scope

The purpose of this note is to exhibit the standard von Neumann measurement construction for the two-qubit singlet within the finite-dimensional geometric reconstruction developed in the surrounding papers. The aim is not to introduce a new measurement Hamiltonian, nor to derive the Born rule independently. Rather, the aim is to show that the canonical quantum-pointer measurement model translates directly into the present framework, with the usual branch separation, pointer readout projectors, and singlet probabilities preserved without distortion.

The system consists of two spin- $\frac{1}{2}$  degrees of freedom prepared in the singlet state

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|+\rangle_z |-\rangle_z - |-\rangle_z |+\rangle_z).$$

The apparatus is represented by two pointer sectors, one coupled to each wing of the Bell experiment. In the finite-dimensional version used here, these pointer sectors are treated as large truncations of continuous position-pointer Hilbert spaces. The total system–apparatus Hilbert space is therefore finite-dimensional,

$$\mathcal{H}_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathcal{K}_A^{(d_A)} \otimes \mathcal{K}_B^{(d_B)}, \quad N = 4d_A d_B,$$

so that the corresponding projective state space is the compact Kähler manifold

$$\mathbb{CP}^{N-1}.$$

The measurement dynamics is generated by the impulsive von Neumann Hamiltonian

$$\hat{H}_{a,b} = \kappa_A(\sigma \cdot a \otimes \mathbb{I}) \otimes \hat{p}_A + \kappa_B(\mathbb{I} \otimes \sigma \cdot b) \otimes \hat{p}_B,$$

where  $a, b \in S^2$  are detector settings,  $\hat{p}_A$  and  $\hat{p}_B$  are pointer momentum operators, and free pointer evolution is neglected during the readout interval. This is the usual idealised strong-measurement limit of von Neumann measurement theory von Neumann [1955]. Through the Ashtekar–Schilling correspondence, the resulting Schrödinger evolution is equivalently Hamiltonian flow on the joint projective manifold Ashtekar and Schilling [1999].

The note therefore has a deliberately limited scope. It establishes a system–apparatus realisation of the singlet measurement inside the finite-dimensional geometric framework. It does not claim a first-principles derivation of the singlet probability kernel, a novel alternative to the von Neumann measurement model, or a Bell-local hidden-variable model.

### 1.2 Relation to Technical Note 2

Technical Note 2 established, non-constructively, that the two-qubit singlet lies within the finite-dimensional Born-weight layer. In that setting, the joint projectors for detector settings  $a, b \in S^2$  are

$$\Pi_{st}(a, b) = |s_a, t_b\rangle\langle s_a, t_b|, \quad s, t \in \{\pm 1\},$$

where  $|s_a\rangle$  is the eigenvector of  $\sigma \cdot a$  with eigenvalue  $s$ , and  $|t_b\rangle$  is the eigenvector of  $\sigma \cdot b$  with eigenvalue  $t$ . The singlet probabilities are then

$$P_{st}(a, b) = \langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle = \frac{1 - st a \cdot b}{4}.$$

Consequently,

$$E(a, b) = \sum_{s, t = \pm 1} st P_{st}(a, b) = -a \cdot b.$$

Technical Note 2 therefore supplies the probability kernel. It does not supply an explicit system–apparatus Hamiltonian whose readout sectors realise that kernel dynamically. The present note fills that constructive slot. It takes the singlet probability law from the upstream finite-dimensional Born-weight layer and shows how it is implemented by a standard quantum-pointer measurement interaction.

This distinction is important. The present note does not attempt to rederive

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}$$

from isotropy, anchor cases, or Fubini–Study cell volumes. Instead, it shows how the probability law already obtained in the finite-dimensional layer appears as the branch-sector weight of an explicit von Neumann readout model.

### 1.3 Contribution of the Present Note

The contribution is one of compatibility and architectural closure. The standard measurement construction is embedded into the finite-dimensional geometric setting without requiring an alternative measurement formalism. In particular:

1. the two measurement couplings are represented by explicit setting-dependent Hamiltonian terms;
2. the system–apparatus evolution separates the four pointer branches associated with outcomes  $s, t \in \{\pm 1\}$ ;
3. pointer-sector projectors define the operational outcome sectors;
4. the branch weights reduce to the standard projector expectations;
5. direct summation yields  $E(a, b) = -a \cdot b$ ;
6. marginal summation gives operational no-signalling.

No Fubini–Study Voronoi-volume construction is used. Outcome sectors are represented by pointer projectors of the form

$$\widehat{M}_{st}^{(a, b)} = \mathbb{I}_{AB} \otimes \widehat{Q}_A^s \otimes \widehat{Q}_B^t,$$

and their weights are computed using the operator-form probability rule

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a, b)} | \Psi_T \rangle.$$

In the interpretive language of the wider framework, these pointer sectors correspond to measurement-context outcome basins. The technical calculation in this note fixes their operational weights. The further ontological question of which sector is realised in a single run is the basin-selection question of the broader programme. The present note assumes the programme-level view that the ontic state is definite and that a realised measurement context selects one outcome sector, while the sector weights are supplied by the finite-dimensional probability layer.

The construction is deterministic per realised context but not Bell-local. It does not define factorised outcome functions

$$R_A(\lambda, a), \quad R_B(\lambda, b).$$

Instead, the relevant outcome structure is context-indexed:

$$F^{(a, b)} : \Sigma_{SA}^{(N)} \longrightarrow \{\pm 1\} \times \{\pm 1\}.$$

This is necessary for consistency with Bell-type constraints Bell [1964], Clauser et al. [1969]. The model is operationally no-signalling because the marginals satisfy

$$P_A(\pm 1 | a, b) = P_B(\pm 1 | a, b) = \frac{1}{2}.$$

Earlier formulations of this construction used projective-cell language. The present formulation instead uses pointer-projector readout, because this is the standard operational representation of a von Neumann measurement and connects directly to the finite-dimensional Born-weight layer used upstream.

## 1.4 Structure of the Paper

Section 2 defines the joint system–apparatus geometry, including the finite-dimensional pointer truncation and the projective state space  $\mathbb{CP}^{N-1}$ . Section 3 introduces the impulsive von Neumann Hamiltonian and proves the factorisation of the measurement unitary in the ideal readout limit. Section 4 derives the branch-separated final state and states the finite-separation error assumptions.

Section 5 defines the pointer-induced outcome sectors and the corresponding operator-form branch weights. It also verifies a weak operational-consistency result for shared pointer channels in compatible contexts. Section 6 evaluates the singlet probabilities and derives

$$E(a, b) = -a \cdot b.$$

Section 7 proves operational no-signalling by direct marginal summation. Section 8 explains the Bell-consistency of the construction, emphasising context-indexed outcome sectors and the absence of a global joint assignment. Section 9 discusses the role of the result as a compatibility and architectural-closure result, including its relation to Bohmian branch selection and to the later Lean formalisation layer.

The appendices record the finite-dimensional pointer-truncation convention, the finite-separation error bounds, and the relation of this note to the surrounding dependency stack.

## 2 Joint System-Apparatus Geometry

### 2.1 Two-Qubit System

The measured system consists of two spin- $\frac{1}{2}$  degrees of freedom. Its Hilbert space is

$$\mathcal{H}_{AB} = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4.$$

For detector settings

$$a, b \in S^2,$$

we write

$$\sigma_a = \sigma \cdot a, \quad \sigma_b = \sigma \cdot b,$$

where

$$\sigma = (\sigma_x, \sigma_y, \sigma_z)$$

denotes the Pauli triple.

For each  $s, t \in \{\pm 1\}$ , let

$$|s_a\rangle$$

be the eigenvector of  $\sigma_a$  with eigenvalue  $s$ , and let

$$|t_b\rangle$$

be the eigenvector of  $\sigma_b$  with eigenvalue  $t$ . The corresponding joint eigenvector is

$$|s_a, t_b\rangle = |s_a\rangle \otimes |t_b\rangle.$$

The associated rank-one projector is

$$\Pi_{st}(a, b) = |s_a, t_b\rangle \langle s_a, t_b|.$$

The singlet state is

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|+\rangle_z |-\rangle_z - |-\rangle_z |+\rangle_z).$$

It is rotationally invariant in the standard sense that for any  $U \in SU(2)$ ,

$$(U \otimes U)|\psi^-\rangle = |\psi^-\rangle$$

up to an irrelevant global phase. This is the symmetry property responsible for the dependence of the final correlation only on

$$a \cdot b.$$

## 2.2 Finite-Dimensional Pointer Sectors

Each wing of the measurement is coupled to a pointer sector. In the ideal continuous model, the pointer Hilbert spaces would be

$$L^2(\mathbb{R})_A, \quad L^2(\mathbb{R})_B.$$

For the finite-dimensional formulation used in this note, we replace these by large finite-dimensional truncations

$$\mathcal{K}_A^{(d_A)}, \quad \mathcal{K}_B^{(d_B)}.$$

These truncations should be understood as finite detector windows with sufficiently many position modes to resolve the two pointer outcomes during the readout interval.

The pointer sectors contain two macroscopically distinct readout subspaces,

$$\mathcal{K}_A^{(+)} \oplus \mathcal{K}_A^{(-)} \subseteq \mathcal{K}_A^{(d_A)},$$

and

$$\mathcal{K}_B^{(+)} \oplus \mathcal{K}_B^{(-)} \subseteq \mathcal{K}_B^{(d_B)}.$$

Let

$$\hat{Q}_A^+, \hat{Q}_A^-$$

be the orthogonal projectors onto the  $A$ -pointer readout subspaces, and let

$$\hat{Q}_B^+, \hat{Q}_B^-$$

be the corresponding projectors for the  $B$ -pointer.

They satisfy

$$\hat{Q}_A^+ + \hat{Q}_A^- = \mathbb{I}_A^P, \quad \hat{Q}_A^+ \hat{Q}_A^- = 0,$$

and

$$\hat{Q}_B^+ + \hat{Q}_B^- = \mathbb{I}_B^P, \quad \hat{Q}_B^+ \hat{Q}_B^- = 0.$$

The finite-dimensional pointer momentum operators

$$\hat{p}_A, \quad \hat{p}_B$$

are self-adjoint generators of pointer translation within the truncated detector window. No exact infinite-dimensional canonical commutation relation is required for the argument. The only required property is that the measurement unitary generated by these operators separates initially localised pointer packets into the corresponding readout subspaces with controllable leakage.

## 2.3 Joint Projective State Space

The full finite-dimensional system-apparatus Hilbert space is

$$\mathcal{H}_{SA}^{(N)} = \mathcal{H}_{AB} \otimes \mathcal{K}_A^{(d_A)} \otimes \mathcal{K}_B^{(d_B)},$$

with

$$N = 4d_A d_B.$$

The associated pure-state ray space is

$$\mathbb{CP}^{N-1}.$$

This projective space is compact and carries the standard Kähler structure

$$(\omega_{FS}, g_{FS}, J_{FS}),$$

where  $\omega_{FS}$  is the Fubini-Study symplectic form,  $g_{FS}$  is the Fubini-Study metric, and  $J_{FS}$  is the compatible complex structure.

In the finite-dimensional geometric formulation, a normalised vector

$$|\Psi\rangle \in \mathcal{H}_{SA}^{(N)}$$

corresponds to the ray

$$[\Psi] \in \mathbb{CP}^{N-1}.$$

Global phases are physically irrelevant:

$$|\Psi\rangle \sim e^{i\chi} |\Psi\rangle.$$

The finite-dimensional system-apparatus state used below is therefore an element of

$$\mathbb{CP}^{N-1},$$

rather than a vector in Hilbert space considered up to no equivalence relation.

## 2.4 Hamiltonian Flow in the Geometric Formulation

A self-adjoint Hamiltonian operator

$$\hat{H} : \mathcal{H}_{SA}^{(N)} \rightarrow \mathcal{H}_{SA}^{(N)}$$

defines the expectation-value function

$$h_{\hat{H}}([\Psi]) = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle}$$

on  $\mathbb{CP}^{N-1}$ .

The Hamiltonian vector field  $X_{h_{\hat{H}}}$  is defined by

$$\iota_{X_{h_{\hat{H}}}} \omega_{FS} = dh_{\hat{H}}.$$

The resulting Hamiltonian flow on  $\mathbb{CP}^{N-1}$  is equivalent to Schrödinger evolution on the underlying Hilbert space. This is the standard Ashtekar-Schilling correspondence between finite-dimensional quantum mechanics and Hamiltonian flow on projective state space Ashtekar and Schilling [1999].

Thus the von Neumann measurement interaction used below may be read in two equivalent ways:

1. as unitary Schrödinger evolution on  $\mathcal{H}_{SA}^{(N)}$ ;
2. as deterministic Hamiltonian flow on the compact Kähler manifold  $\mathbb{CP}^{N-1}$ .

This equivalence is the geometric route by which the standard quantum-pointer measurement model is embedded into the present finite-dimensional framework.

## 2.5 Finite-Truncation Scope

All exact statements in this note are finite-dimensional statements at fixed

$$d_A, d_B < \infty.$$

The continuous-pointer model is recovered only as a controlled comparison limit,

$$d_A, d_B \rightarrow \infty,$$

and is not required for the finite-dimensional construction.

The truncation must satisfy three operational conditions:

1. the initial pointer states are well localised within the finite detector window;
2. the measurement coupling separates the shifted pointer packets into disjoint readout subspaces;
3. boundary or wrap-around effects remain negligible during the readout interval.

These conditions are collected later into the branch-separation parameters

$$\epsilon_A, \quad \epsilon_B.$$

The strong-readout limit corresponds to

$$\epsilon_A, \epsilon_B \rightarrow 0.$$

At finite separation, all probabilities and correlations acquire corrections of order

$$O(\epsilon_A + \epsilon_B).$$

This finite-truncation framing keeps the construction within compact finite-dimensional projective geometry while preserving the operational content of the usual continuous von Neumann pointer model.



### 3 Von Neumann Measurement Hamiltonian

#### 3.1 Detector Settings and Spin Operators

Let

$$a, b \in S^2$$

denote the detector settings for the two wings of the Bell experiment. Define the corresponding spin observables

$$\sigma_a = \sigma \cdot a, \quad \sigma_b = \sigma \cdot b,$$

where

$$\sigma = (\sigma_x, \sigma_y, \sigma_z)$$

is the Pauli triple.

The eigenvalue equations are

$$\sigma_a |s_a\rangle = s |s_a\rangle, \quad \sigma_b |t_b\rangle = t |t_b\rangle,$$

with

$$s, t \in \{\pm 1\}.$$

The joint eigenvectors

$$|s_a, t_b\rangle = |s_a\rangle \otimes |t_b\rangle$$

form an orthonormal basis of

$$\mathcal{H}_{AB} = \mathbb{C}^2 \otimes \mathbb{C}^2.$$

The corresponding joint projectors are

$$\Pi_{st}(a, b) = |s_a, t_b\rangle \langle s_a, t_b|.$$

Equivalently,

$$\Pi_{st}(a, b) = \Pi_s^A(a) \otimes \Pi_t^B(b),$$

where

$$\Pi_s^A(a) = \frac{1}{2} (\mathbb{I} + s \sigma_a), \quad \Pi_t^B(b) = \frac{1}{2} (\mathbb{I} + t \sigma_b).$$

#### 3.2 Impulsive Readout Hamiltonian

The measurement interaction is taken to be the standard von Neumann pointer coupling. In the main construction we work in the impulsive-readout idealisation, where free pointer evolution is negligible during the measurement interval. Thus we set

$$\hat{H}_{\text{free}} = 0$$

in the primary derivation.

The readout Hamiltonian is

$$\hat{H}_{a,b} = \hat{H}_A(a) + \hat{H}_B(b),$$

with

$$\hat{H}_A(a) = \kappa_A (\sigma_a \otimes \mathbb{I}) \otimes \hat{p}_A \otimes \mathbb{I}_B^P,$$

and

$$\hat{H}_B(b) = \kappa_B (\mathbb{I} \otimes \sigma_b) \otimes \mathbb{I}_A^P \otimes \hat{p}_B.$$

Here

$$\hat{p}_A : \mathcal{K}_A^{(d_A)} \rightarrow \mathcal{K}_A^{(d_A)}, \quad \hat{p}_B : \mathcal{K}_B^{(d_B)} \rightarrow \mathcal{K}_B^{(d_B)}$$

are finite-dimensional pointer-momentum operators, and

$$\kappa_A, \kappa_B > 0$$

are coupling constants.

The Hamiltonian acts on

$$\mathcal{H}_{SA}^{(N)} = \mathcal{H}_{AB} \otimes \mathcal{K}_A^{(d_A)} \otimes \mathcal{K}_B^{(d_B)}.$$

The physical role of  $\hat{H}_A(a)$  is to translate the  $A$ -pointer conditional on the eigenvalue of  $\sigma_a$ . Similarly,  $\hat{H}_B(b)$  translates the  $B$ -pointer conditional on the eigenvalue of  $\sigma_b$ .

### 3.3 Commutation and Unitary Factorisation

The two interaction terms commute:

$$[\hat{H}_A(a), \hat{H}_B(b)] = 0.$$

This follows because

$$[\sigma_a \otimes \mathbb{I}, \mathbb{I} \otimes \sigma_b] = 0$$

on the two-qubit system, while

$$[\hat{p}_A \otimes \mathbb{I}_B^P, \mathbb{I}_A^P \otimes \hat{p}_B] = 0$$

on the two pointer sectors.

The measurement unitary over readout time  $T$  is therefore

$$\hat{U}_{a,b}(T) = \exp\left(-\frac{iT}{\hbar} \hat{H}_{a,b}\right).$$

Using commutativity, this factorises as

$$\hat{U}_{a,b}(T) = \hat{U}_A(a; T) \hat{U}_B(b; T),$$

where

$$\hat{U}_A(a; T) = \exp\left(-\frac{i}{\hbar} \kappa_A T (\sigma_a \otimes \mathbb{I}) \otimes \hat{p}_A \otimes \mathbb{I}_B^P\right),$$

and

$$\hat{U}_B(b; T) = \exp\left(-\frac{i}{\hbar} \kappa_B T (\mathbb{I} \otimes \sigma_b) \otimes \mathbb{I}_A^P \otimes \hat{p}_B\right).$$

For a system eigenstate

$$|s_a, t_b\rangle,$$

the unitary acts on the pointer factors as

$$\hat{U}_{a,b}(T) (|s_a, t_b\rangle \otimes |\phi_A^0\rangle \otimes |\phi_B^0\rangle) = |s_a, t_b\rangle \otimes e^{-is\kappa_A T \hat{p}_A / \hbar} |\phi_A^0\rangle \otimes e^{-it\kappa_B T \hat{p}_B / \hbar} |\phi_B^0\rangle.$$

Thus each branch receives a pointer translation determined by the corresponding pair of eigenvalues

$$(s, t).$$

### 3.4 Parameter Locality at the Hamiltonian-Control Level

The readout Hamiltonian has local parameter dependence:

$$\frac{\partial \hat{H}_A(a)}{\partial b} = 0, \quad \frac{\partial \hat{H}_B(b)}{\partial a} = 0.$$

The setting  $a$  enters only the  $A$ -wing coupling, and the setting  $b$  enters only the  $B$ -wing coupling.

This statement is deliberately limited. It is parameter locality at the level of the Hamiltonian controls. It is not Bell-local factorisation of outcomes. The final outcome sectors for a Bell experiment are context-indexed by the pair

$$(a, b),$$

and the Bell-consistency issue is treated separately below.

This distinction is necessary because any deterministic single-outcome model reproducing the singlet correlation

$$E(a, b) = -a \cdot b$$

cannot also satisfy Bell-local outcome factorisation under the usual measurement-independence assumptions Bell [1964], Clauser et al. [1969].

### 3.5 Free Evolution and the Impulsive Limit

The main construction sets

$$\hat{H}_{\text{free}} = 0$$

during the readout interval. This is the standard impulsive-measurement idealisation: the interaction is assumed to dominate the pointer free evolution over the finite time  $T$ .

If one includes finite pointer kinetic terms, for example

$$\hat{H}_{\text{free}} = \frac{\hat{p}_A^2}{2m_A} + \frac{\hat{p}_B^2}{2m_B},$$

then pointer spreading introduces additional finite-time corrections. These corrections are not part of the ideal derivation. They may be absorbed into the branch-leakage parameters

$$\epsilon_A, \quad \epsilon_B,$$

introduced in Section 4.

The exact result is therefore obtained in the strong impulsive-readout limit:

$$\epsilon_A, \epsilon_B \rightarrow 0.$$

At finite pointer separation, all branch probabilities and correlations acquire corrections of order

$$O(\epsilon_A + \epsilon_B).$$

The role of the impulsive limit is not to alter the singlet probability law. Its role is to make the pointer readout sectors sharply separated, so that the four measurement branches can be treated as operationally distinct outcome sectors.

## 4 Branch Separation Under Hamiltonian Flow

### 4.1 Initial Singlet-Pointer State

The initial system-apparatus state is taken to be

$$|\Psi_0\rangle = |\psi^-\rangle \otimes |\phi_A^0\rangle \otimes |\phi_B^0\rangle,$$

where

$$|\psi^-\rangle \in \mathcal{H}_{AB}$$

is the two-qubit singlet, and

$$|\phi_A^0\rangle \in \mathcal{K}_A^{(d_A)}, \quad |\phi_B^0\rangle \in \mathcal{K}_B^{(d_B)}$$

are initially localised pointer states.

The pointer states are assumed to be initially localised near the origin of their respective readout axes before the measurement interaction. After the impulsive coupling, the pointer packets are translated into one of two macroscopically distinguishable readout sectors,

$$\mathcal{K}_A^{(+)}, \mathcal{K}_A^{(-)}$$

for the  $A$ -pointer and

$$\mathcal{K}_B^{(+)}, \mathcal{K}_B^{(-)}$$

for the  $B$ -pointer.

The ideal measurement limit is the regime in which these translated pointer packets have negligible overlap with the wrong readout sectors.

### 4.2 Expansion in the Joint Measurement Basis

For fixed detector settings

$$a, b \in S^2,$$

the joint measurement basis is

$$\{|s_a, t_b\rangle : s, t \in \{\pm 1\}\}.$$

The singlet state can be expanded in this basis as

$$|\psi^-\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a,b) |s_a, t_b\rangle,$$

where

$$c_{st}(a,b) = \langle s_a, t_b | \psi^-\rangle.$$

Therefore the initial system-apparatus state may be written as

$$|\Psi_0\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a,b) |s_a, t_b\rangle \otimes |\phi_A^0\rangle \otimes |\phi_B^0\rangle.$$

This expansion is purely a change of basis in the two-qubit system. No probability rule has yet been applied.

### 4.3 Pointer Translation by the Measurement Interaction

From Section 3, the measurement unitary factorises as

$$\hat{U}_{a,b}(T) = \hat{U}_A(a; T) \hat{U}_B(b; T).$$

For each branch

$$|s_a, t_b\rangle,$$

the  $A$ -pointer is translated according to the eigenvalue  $s$ , and the  $B$ -pointer is translated according to the eigenvalue  $t$ .

Define

$$|\phi_A^s(T)\rangle = \exp\left(-\frac{i}{\hbar} s \kappa_A T \hat{p}_A\right) |\phi_A^0\rangle,$$

and

$$|\phi_B^t(T)\rangle = \exp\left(-\frac{i}{\hbar} t \kappa_B T \hat{p}_B\right) |\phi_B^0\rangle.$$

Then

$$\hat{U}_{a,b}(T) (|s_a, t_b\rangle \otimes |\phi_A^0\rangle \otimes |\phi_B^0\rangle) = |s_a, t_b\rangle \otimes |\phi_A^s(T)\rangle \otimes |\phi_B^t(T)\rangle.$$

Thus the measurement Hamiltonian correlates each system eigenbranch with a corresponding pair of pointer states.

### 4.4 Branch-Separated Final State

Applying the measurement unitary to the initial state gives

$$|\Psi_T\rangle = \hat{U}_{a,b}(T) |\Psi_0\rangle.$$

Using linearity,

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a,b) |s_a, t_b\rangle \otimes |\phi_A^s(T)\rangle \otimes |\phi_B^t(T)\rangle.$$

This is the standard von Neumann branch-separated form. Each term in the sum consists of:

1. a two-qubit system eigenbranch  $|s_a, t_b\rangle$ ;
2. an  $A$ -pointer state shifted toward readout sector  $s$ ;
3. a  $B$ -pointer state shifted toward readout sector  $t$ .

The four branch rays are

$$[|s_a, t_b\rangle \otimes |\phi_A^s(T)\rangle \otimes |\phi_B^t(T)\rangle], \quad s, t \in \{\pm 1\}.$$

In the strong-readout regime these four branches are operationally distinguishable because their pointer supports lie in disjoint readout sectors.

#### 4.5 Asymptotic Branch-Separation Lemma

Let

$$\hat{Q}_A^s$$

be the projector onto the  $A$ -pointer readout sector corresponding to outcome  $s$ , and let

$$\hat{Q}_B^t$$

be the corresponding projector for the  $B$ -pointer.

We assume that the pointer coupling and readout time are chosen so that, for all  $s, t \in \{\pm 1\}$ ,

$$\langle \phi_A^s(T) | \hat{Q}_A^s | \phi_A^s(T) \rangle \geq 1 - \epsilon_A,$$

and

$$\langle \phi_A^s(T) | \hat{Q}_A^{-s} | \phi_A^s(T) \rangle \leq \epsilon_A,$$

with analogous bounds

$$\langle \phi_B^t(T) | \hat{Q}_B^t | \phi_B^t(T) \rangle \geq 1 - \epsilon_B,$$

and

$$\langle \phi_B^t(T) | \hat{Q}_B^{-t} | \phi_B^t(T) \rangle \leq \epsilon_B.$$

Here

$$\epsilon_A, \epsilon_B \geq 0$$

measure pointer leakage into the wrong readout sector.

**Lemma.** Under the above branch-separation assumptions, the post-measurement state

$$|\Psi_T\rangle = \sum_{s,t} c_{st}(a,b) |s_a, t_b\rangle \otimes |\phi_A^s(T)\rangle \otimes |\phi_B^t(T)\rangle$$

has four operationally distinguishable pointer branches up to total leakage of order

$$O(\epsilon_A + \epsilon_B).$$

**Proof.** The branch associated with  $(s, t)$  has pointer support concentrated in

$$\mathcal{K}_A^{(s)} \otimes \mathcal{K}_B^{(t)}.$$

The probability that the  $A$ -pointer lies outside  $\mathcal{K}_A^{(s)}$  is at most  $\epsilon_A$ , and the probability that the  $B$ -pointer lies outside  $\mathcal{K}_B^{(t)}$  is at most  $\epsilon_B$ . Hence the total probability that at least one pointer is found outside its intended readout sector is bounded by

$$\epsilon_A + \epsilon_B$$

up to higher-order overlap terms. Orthogonality of the system eigenvectors

$$|s_a, t_b\rangle$$

ensures that different system branches do not contribute cross terms in the pointer-sector probability calculation. Therefore the four pointer branches are distinguishable up to leakage of order

$$O(\epsilon_A + \epsilon_B).$$

□

#### 4.6 Finite-Separation Error Terms

In the ideal strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

the pointer sectors are perfectly separated:

$$\langle \phi_A^s(T) | \hat{Q}_A^s | \phi_A^s(T) \rangle \rightarrow 1, \quad \langle \phi_A^s(T) | \hat{Q}_A^{-s} | \phi_A^s(T) \rangle \rightarrow 0,$$

and similarly for  $B$ .

At finite separation, branch weights differ from their ideal values by terms controlled by

$$\epsilon_A + \epsilon_B.$$

Consequently, all later probability and correlation statements have the form

$$P_{st}(a, b; T) = P_{st}^{\text{ideal}}(a, b) + O(\epsilon_A + \epsilon_B),$$

and

$$E_T(a, b) = E_{\text{ideal}}(a, b) + O(\epsilon_A + \epsilon_B).$$

The ideal limit is therefore not a separate physical postulate. It is the controlled regime in which the pointer branches become operationally distinguishable and the measurement interaction realises the projective singlet measurement to arbitrary accuracy.

### 5 Pointer-Induced Outcome Sectors

#### 5.1 Pointer Readout Projectors

The operational readout of the apparatus is represented by pointer-sector projectors. For each outcome

$$s \in \{\pm 1\},$$

let

$$\hat{Q}_A^s$$

denote the projector onto the  $A$ -pointer readout sector associated with outcome  $s$ . Similarly, for

$$t \in \{\pm 1\},$$

let

$$\hat{Q}_B^t$$

denote the projector onto the  $B$ -pointer readout sector associated with outcome  $t$ .

These projectors satisfy

$$\hat{Q}_A^+ + \hat{Q}_A^- = \mathbb{I}_A^P, \quad \hat{Q}_A^+ \hat{Q}_A^- = 0,$$

and

$$\hat{Q}_B^+ + \hat{Q}_B^- = \mathbb{I}_B^P, \quad \hat{Q}_B^+ \hat{Q}_B^- = 0.$$

The joint pointer-sector projector for outcome pair  $(s, t)$  is

$$\hat{M}_{st} = \mathbb{I}_{AB} \otimes \hat{Q}_A^s \otimes \hat{Q}_B^t.$$

Although the operator  $\hat{M}_{st}$  acts only on the pointer readout sectors, the realised measurement context is still labelled by

$$(a, b),$$

because the Hamiltonian that generated the branch-separated state depends on those detector settings.

Thus, when necessary, we write

$$\hat{M}_{st}^{(a,b)}$$

to emphasise that the same pointer readout sector is being evaluated after the context-dependent measurement interaction

$$\hat{U}_{a,b}(T).$$

## 5.2 Outcome Sectors for a Measurement Context

For a fixed context

$$(a, b),$$

the post-measurement state is

$$|\Psi_T\rangle = \sum_{u,v \in \{\pm 1\}} c_{uv}(a, b) |u_a, v_b\rangle \otimes |\phi_A^u(T)\rangle \otimes |\phi_B^v(T)\rangle.$$

The outcome sector  $(s, t)$  is defined operationally by the pointer projector

$$\widehat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \widehat{Q}_A^s \otimes \widehat{Q}_B^t.$$

The corresponding branch weight is

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

This is the standard operator-form probability for the pointer readout. In the present framework, it is also the operational weight assigned to the outcome sector selected by the measurement context.

The four projectors form a complete orthogonal resolution of the pointer readout space:

$$\sum_{s,t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \mathbb{I}_A^P \otimes \mathbb{I}_B^P,$$

and

$$\widehat{M}_{st}^{(a,b)} \widehat{M}_{s't'}^{(a,b)} = 0$$

whenever

$$(s, t) \neq (s', t').$$

Therefore the four outcome sectors form a projective measurement context on the system-apparatus state after readout.

## 5.3 Operator-Form Branch Weights

Substituting the branch-separated state gives

$$P_{st}(a, b; T) = \left\langle \sum_{u,v} c_{uv} |u_a, v_b\rangle |\phi_A^u(T)\rangle |\phi_B^v(T)\rangle \left| \widehat{M}_{st}^{(a,b)} \right| \sum_{u',v'} c_{u'v'} |u'_a, v'_b\rangle |\phi_A^{u'}(T)\rangle |\phi_B^{v'}(T)\rangle \right\rangle.$$

Because the system vectors

$$|u_a, v_b\rangle$$

form an orthonormal basis, cross terms vanish:

$$\langle u_a, v_b | u'_a, v'_b \rangle = \delta_{uu'} \delta_{vv'}.$$

Thus

$$P_{st}(a, b; T) = \sum_{u,v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \langle \phi_A^u(T) | \widehat{Q}_A^s | \phi_A^u(T) \rangle \langle \phi_B^v(T) | \widehat{Q}_B^t | \phi_B^v(T) \rangle.$$

Using the branch-separation assumptions from Section 4, the term with

$$u = s, \quad v = t$$

contributes

$$|c_{st}(a, b)|^2 (1 + O(\epsilon_A + \epsilon_B)),$$

while the remaining terms contribute only leakage of order

$$O(\epsilon_A + \epsilon_B).$$

Hence

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + O(\epsilon_A + \epsilon_B).$$

A fully explicit bound on the wrong-branch leakage terms is given in Appendix B.2.

In the strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

we obtain

$$P_{st}(a, b) = |c_{st}(a, b)|^2.$$

This is the point at which the measurement Hamiltonian connects the branch-separated apparatus description to the standard singlet probabilities evaluated in Section 6.

#### 5.4 CSD Basin Interpretation

The preceding calculation is operational and uses only standard pointer projectors. The CSD interpretation adds the following reading.

Let

$$\pi_{SA}^{(N)} : \Sigma_{SA}^{(N)} \rightarrow \mathbb{CP}^{N-1}$$

be the system-apparatus projection in the finite-dimensional quantum-effective sector. For each measurement context  $(a, b)$ , the pointer-sector projector

$$\widehat{M}_{st}^{(a,b)}$$

defines an epistemic outcome sector

$$\Omega_{st}^{\text{ep}}(a, b) \subset \mathbb{CP}^{N-1}.$$

The corresponding ontic basin is the inverse image

$$\Omega_{st}^{\Sigma}(a, b) = \left( \pi_{SA}^{(N)} \right)^{-1} \left( \Omega_{st}^{\text{ep}}(a, b) \right).$$

This basin notation should not be read as an additional probability calculation. The branch weight is computed by the operator-form expression

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

The role of the basin language is interpretive and structural. It identifies the outcome sector in the ontic framework corresponding to the operational pointer reading.

The further question of which outcome sector is realised in a single run is the basin-selection question of the wider programme. The present note does not independently derive that selection rule. It assumes the programme-level interpretation that the ontic state is definite and lies in one realised outcome sector, while the technical result established here concerns the system–apparatus Hamiltonian and the operational weights of the resulting pointer sectors. A fuller treatment of basin selection should be tracked as a programme-level item rather than as a claim of the present note.

A further distinction is important. In this subsection, the term “basin” denotes the pullback of an operational pointer sector under

$$\pi_{SA}^{(N)}.$$

It does not denote an independently classified attractor basin or a dynamically derived family of all physically realisable measurement partitions. The latter problem is the dynamics-based classification problem for measurement partitions, tracked in the programme dependency map as G3b. The present note supplies one explicit quantum-pointer construction and its associated pullback sectors. It does not claim a general classification theorem for all apparatus Hamiltonians or all admissible outcome partitions.

#### 5.5 No Use of Fubini-Study Voronoi Volumes

No Fubini-Study Voronoi-volume construction is used in this note.

The outcome sectors are not defined as nearest-ray cells in projective space. They are defined by physical pointer readout projectors:

$$\widehat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \widehat{Q}_A^s \otimes \widehat{Q}_B^t.$$

Their weights are not computed by geometric cell volumes. They are computed by the finite-dimensional operator-form rule

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$



This distinction is important. The present construction is a quantum-pointer Hamiltonian realisation of the singlet measurement, not a Voronoi partition argument. The result depends on branch separation under the measurement interaction and on the standard projector probabilities for pointer sectors.

## 5.6 Weak Operational Consistency for Shared Pointer Channels

Technical Note 2 records an operational consistency condition, H5, according to which the weight assigned to a fixed macroscopic outcome channel should not depend on the compatible orthogonal completion in which that channel appears. The present note does not prove the full Gleason-style version of that condition. It does, however, verify the weak form needed for the present two-wing pointer construction.

Fix the  $A$ -wing detector setting  $a$ , and consider two contexts

$$(a, b), \quad (a, b'),$$

which share the same  $A$ -wing coupling

$$\hat{H}_A(a) = \kappa_A(\sigma_a \otimes \mathbb{I}) \otimes \hat{p}_A \otimes \mathbb{I}_B^P,$$

but differ in the  $B$ -wing coupling. The  $A$ -wing pointer-channel projector for outcome  $s \in \{\pm 1\}$  is

$$\hat{M}_A^s = \mathbb{I}_{AB} \otimes \hat{Q}_A^s \otimes \mathbb{I}_B^P.$$

Equivalently,

$$\hat{M}_A^s = \sum_{t \in \{\pm 1\}} \hat{M}_{st}^{(a,b)} = \sum_{t \in \{\pm 1\}} \hat{M}_{st}^{(a,b')}.$$

Thus the operator representing the  $A$ -wing macroscopic readout channel is identical in the two compatible contexts.

Let

$$\Omega_{A,s}^{\text{ep}} \subset \mathbb{CP}^{N-1}$$

denote the epistemic pointer sector associated with  $\hat{M}_A^s$ , and define the corresponding pullback sector by

$$\Omega_{A,s}^\Sigma = \left( \pi_{SA}^{(N)} \right)^{-1} \left( \Omega_{A,s}^{\text{ep}} \right).$$

Since the projector  $\hat{M}_A^s$  is the same in the two contexts, the associated epistemic sector is the same, and therefore the pullback sector is the same:

$$\Omega_{A,s}^\Sigma(a, b) = \Omega_{A,s}^\Sigma(a, b')$$

up to the common null-boundary convention used for pointer readout sectors.

This establishes a weak form of the H5 consistency condition for the present von Neumann pointer model: when a macroscopic pointer channel is literally represented by the same tensor-factor readout projector in two compatible contexts, the associated CSD pullback sector is independent of the remote compatible completion.

This result should not be read as a full proof of Gleason-style operational consistency for all orthogonal completions. It applies only to shared pointer channels of the specific system–apparatus construction used here. The stronger question, namely which apparatus Hamiltonians generate completion-independent outcome sectors in full generality, remains outside the scope of the present note.

## 6 Singlet Probabilities and Correlation

### 6.1 Projector Probabilities from the Singlet State

The preceding section showed that, in the strong-readout limit, the pointer-sector probability is

$$P_{st}(a, b) = |c_{st}(a, b)|^2,$$

where

$$c_{st}(a, b) = \langle s_a, t_b | \psi^- \rangle.$$

Equivalently, let

$$\Pi_{st}(a, b) = |s_a, t_b\rangle \langle s_a, t_b|$$

be the joint rank-one projector associated with the spin outcomes

$$s, t \in \{\pm 1\}.$$

Then

$$|c_{st}(a, b)|^2 = |\langle s_a, t_b | \psi^- \rangle|^2 = \langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle.$$

This is the bridge between the branch amplitudes appearing in the system–apparatus state and the operator-form probabilities computed in Technical Note 2.

Using

$$\Pi_s^A(a) = \frac{1}{2} (\mathbb{I} + s \sigma_a), \quad \Pi_t^B(b) = \frac{1}{2} (\mathbb{I} + t \sigma_b),$$

the joint projector is

$$\Pi_{st}(a, b) = \Pi_s^A(a) \otimes \Pi_t^B(b).$$

Therefore

$$\Pi_{st}(a, b) = \frac{1}{4} (\mathbb{I} + s \sigma_a) \otimes (\mathbb{I} + t \sigma_b).$$

The singlet state satisfies

$$\langle \psi^- | \sigma_a \otimes \mathbb{I} | \psi^- \rangle = 0,$$

$$\langle \psi^- | \mathbb{I} \otimes \sigma_b | \psi^- \rangle = 0,$$

and

$$\langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle = -a \cdot b.$$

These are the standard singlet identities.

Substituting into the projector expectation gives

$$\langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle = \frac{1}{4} (1 + st \langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle).$$

Hence

$$\langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle = \frac{1}{4} (1 - st a \cdot b).$$

Thus, in the strong-readout limit,

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

## 6.2 Relation Between Amplitudes and Projector Expectations

The equality

$$|c_{st}(a, b)|^2 = \langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle$$

is used only as a Hilbert-space identity inside the finite-dimensional operational layer.

The coefficient

$$c_{st}(a, b)$$

is defined by expanding the singlet in the joint eigenbasis:

$$|\psi^- \rangle = \sum_{s, t \in \{\pm 1\}} c_{st}(a, b) |s_a, t_b \rangle.$$

Since

$$\Pi_{st}(a, b) = |s_a, t_b \rangle \langle s_a, t_b|,$$

we have

$$\Pi_{st}(a, b) |\psi^- \rangle = c_{st}(a, b) |s_a, t_b \rangle.$$

Therefore

$$\langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle = |c_{st}(a, b)|^2.$$

The role of Technical Note 2 is to supply the upstream evaluation of this projector expectation within the finite-dimensional Born-weight layer. The role of the present note is to show how the same quantities arise as pointer-sector branch weights under an explicit system–apparatus Hamiltonian.

### 6.3 Evaluation of $P_{st}(a, b)$

Let

$$\theta = \arccos(a \cdot b).$$

Then

$$a \cdot b = \cos \theta.$$

The joint probabilities are

$$P_{st}(a, b) = \frac{1 - st \cos \theta}{4}.$$

For equal signs,

$$st = +1,$$

so

$$P_{++}(a, b) = P_{--}(a, b) = \frac{1 - \cos \theta}{4}.$$

Using

$$1 - \cos \theta = 2 \sin^2 \frac{\theta}{2},$$

this becomes

$$P_{++}(a, b) = P_{--}(a, b) = \frac{1}{2} \sin^2 \frac{\theta}{2}.$$

For opposite signs,

$$st = -1,$$

so

$$P_{+-}(a, b) = P_{-+}(a, b) = \frac{1 + \cos \theta}{4}.$$

Using

$$1 + \cos \theta = 2 \cos^2 \frac{\theta}{2},$$

this becomes

$$P_{+-}(a, b) = P_{-+}(a, b) = \frac{1}{2} \cos^2 \frac{\theta}{2}.$$

The four probabilities sum to one:

$$P_{++} + P_{--} + P_{+-} + P_{-+} = 1.$$

These are precisely the Bell-singlet joint probabilities for spin measurements along directions  $a$  and  $b$ .

### 6.4 Derivation of $E(a, b) = -a \cdot b$

The correlation function is

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b).$$

Using the explicit probabilities,

$$E(a, b) = P_{++} + P_{--} - P_{+-} - P_{-+}.$$

Substituting the half-angle expressions gives

$$E(a, b) = \frac{1}{2} \sin^2 \frac{\theta}{2} + \frac{1}{2} \sin^2 \frac{\theta}{2} - \frac{1}{2} \cos^2 \frac{\theta}{2} - \frac{1}{2} \cos^2 \frac{\theta}{2}.$$

Therefore

$$E(a, b) = \sin^2 \frac{\theta}{2} - \cos^2 \frac{\theta}{2}.$$

Since

$$\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2},$$

we obtain

$$E(a, b) = -\cos \theta.$$

Thus

$$E(a, b) = -a \cdot b.$$

This derivation does not rely on an isotropy-to-linearity argument. The angular dependence enters through the standard singlet projector expectation

$$\langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle = \frac{1 - st a \cdot b}{4},$$

and the present note shows how those projector weights are realised as pointer-sector branch weights by the measurement Hamiltonian.

## 6.5 Finite-Separation Corrections

At finite pointer separation, Section 5 gives

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + O(\epsilon_A + \epsilon_B).$$

Using

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4},$$

we have

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + O(\epsilon_A + \epsilon_B).$$

The finite-time correlation is therefore

$$E_T(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b; T).$$

Hence

$$E_T(a, b) = -a \cdot b + O(\epsilon_A + \epsilon_B).$$

In the strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

the exact singlet correlation is recovered:

$$E_T(a, b) \rightarrow -a \cdot b.$$

Thus the finite-dimensional quantum-pointer construction reproduces the Bell-singlet correlation exactly in the ideal measurement limit and approximately, with controlled error, at finite pointer separation.

## 7 Operational No-Signalling

### 7.1 Marginal Probabilities at A

The operational marginal probability for obtaining outcome

$$s \in \{\pm 1\}$$

at wing A is obtained by summing over the unobserved outcome at wing B:

$$P_A(s | a, b) = \sum_{t \in \{\pm 1\}} P_{st}(a, b).$$

Using the strong-readout probabilities derived in Section 6,

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

we have

$$P_A(s | a, b) = \frac{1 - s(+1)a \cdot b}{4} + \frac{1 - s(-1)a \cdot b}{4}.$$

Therefore

$$P_A(s | a, b) = \frac{1 - sa \cdot b}{4} + \frac{1 + sa \cdot b}{4} = \frac{1}{2}.$$

Thus

$$P_A(+1 | a, b) = P_A(-1 | a, b) = \frac{1}{2}.$$

In particular,

$$P_A(s | a, b)$$

is independent of the remote detector setting  $b$ .

## 7.2 Marginal Probabilities at $B$

Similarly, the operational marginal probability for obtaining outcome

$$t \in \{\pm 1\}$$

at wing  $B$  is

$$P_B(t | a, b) = \sum_{s \in \{\pm 1\}} P_{st}(a, b).$$

Using again

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

we find

$$P_B(t | a, b) = \frac{1 - (+1)t a \cdot b}{4} + \frac{1 - (-1)t a \cdot b}{4}.$$

Hence

$$P_B(t | a, b) = \frac{1 - ta \cdot b}{4} + \frac{1 + ta \cdot b}{4} = \frac{1}{2}.$$

Therefore

$$P_B(+1 | a, b) = P_B(-1 | a, b) = \frac{1}{2},$$

independent of the remote detector setting  $a$ .

## 7.3 No-Signalling in the Strong-Readout Limit

The preceding calculations show that, in the strong-readout limit,

$$P_A(s | a, b) = P_A(s | a) = \frac{1}{2},$$

and

$$P_B(t | a, b) = P_B(t | b) = \frac{1}{2}.$$

The local outcome statistics at  $A$  therefore do not depend on the setting chosen at  $B$ , and the local outcome statistics at  $B$  do not depend on the setting chosen at  $A$ . This is operational no-signalling.

The result follows directly from the explicit pointer-sector probabilities:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

No separate no-signalling postulate is introduced. The cancellation of the remote-setting dependence occurs when the unobserved remote outcome is summed over.

## 7.4 Finite-Separation Corrections to Marginals

At finite pointer separation,

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + \Delta_{st}(a, b; T),$$

where

$$\Delta_{st}(a, b; T) = O(\epsilon_A + \epsilon_B).$$

The marginal at  $A$  is then

$$P_A(s \mid a, b; T) = \sum_{t \in \{\pm 1\}} P_{st}(a, b; T).$$

Substituting the finite-separation form gives

$$P_A(s \mid a, b; T) = \frac{1}{2} + \sum_{t \in \{\pm 1\}} \Delta_{st}(a, b; T).$$

Therefore

$$P_A(s \mid a, b; T) = \frac{1}{2} + O(\epsilon_A + \epsilon_B).$$

Similarly,

$$P_B(t \mid a, b; T) = \frac{1}{2} + O(\epsilon_A + \epsilon_B).$$

Thus operational no-signalling is exact in the ideal strong-readout limit and stable under finite pointer-separation errors, with deviations controlled by the same leakage parameters that control branch distinguishability.

## 7.5 Interpretation

The no-signalling result is operational. It concerns the observable statistics available to agents at either wing. It does not require Bell-local outcome factorisation at the ontic level.

In the present construction, the measurement context is labelled by the pair

$$(a, b),$$

and the branch-separated outcome sectors are defined relative to that context. Nevertheless, once the remote outcome is ignored, the local marginal statistics are independent of the remote setting. Hence the model is context-indexed at the level of outcome-sector structure but no-signalling at the level of operational probabilities.

This is the required compatibility condition for a deterministic single-outcome reconstruction of Bell-singlet statistics.

## 8 Bell Consistency

### 8.1 Context-Indexed Outcome Sectors

The construction above is deterministic relative to a realised measurement context. For each pair of detector settings

$$(a, b),$$

the measurement Hamiltonian

$$\hat{H}_{a,b}$$

generates a branch-separated system–apparatus state, and the pointer projectors

$$\hat{M}_{st}^{(a,b)}$$

define the outcome sectors for that context.

In the ontic notation of the broader framework, this corresponds to a context-indexed outcome map

$$F^{(a,b)} : \Sigma_{SA}^{(N)} \longrightarrow \{\pm 1\} \times \{\pm 1\}.$$

At the programme-level interpretive layer, each realised context is represented by a definite outcome-sector assignment

$$F^{(a,b)}(x) = (s, t),$$

for the underlying ontic state  $x$ . The present note does not independently derive this basin-selection map; it constructs the corresponding system–apparatus pointer sectors and their operational weights.

The dependence on the full context  $(a, b)$  is essential. The model does not assign outcomes through context-independent local functions. Instead, each experimental arrangement determines its own system–apparatus Hamiltonian and its own pointer-sector decomposition.

This is the standard structure of a contextual measurement model. Outcomes are definite within a performed context, but there is no assumption that all outcomes for all incompatible contexts are simultaneously defined.

## 8.2 Absence of Bell-Local Factorisation

Bell-local deterministic hidden-variable models assume outcome functions of the form

$$A(a, \lambda) \in \{\pm 1\}, \quad B(b, \lambda) \in \{\pm 1\},$$

where  $\lambda$  denotes the hidden state, and where the local outcome at  $A$  depends only on  $a$  and  $\lambda$ , while the local outcome at  $B$  depends only on  $b$  and  $\lambda$ .

Equivalently, the joint outcome assignment factorises as

$$F^{(a,b)}(\lambda) = (A(a, \lambda), B(b, \lambda)).$$

The present construction does not assume such factorisation. The outcome sectors are defined by the full measurement context:

$$F^{(a,b)} : \Sigma_{SA}^{(N)} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

The notation is not cosmetic. It records that the relevant sector decomposition is induced by the full system–apparatus Hamiltonian associated with the experimental context  $(a, b)$ .

This is necessary for consistency with Bell’s theorem Bell [1964]. A deterministic single-outcome model reproducing

$$E(a, b) = -a \cdot b$$

cannot also satisfy Bell-local outcome factorisation under the usual measurement-independence assumptions.

Thus the construction is deterministic, but not Bell-local in the factorised hidden-variable sense.

## 8.3 No Global Joint Assignment

The same point can be expressed in Fine’s language. Consider four CHSH settings

$$a, \quad a', \quad b, \quad b'.$$

A Bell-local deterministic model would assign, for each hidden state  $\lambda$ , a quadruple

$$(A(a, \lambda), A(a', \lambda), B(b, \lambda), B(b', \lambda)).$$

Such a quadruple defines a single joint assignment for all four incompatible measurement settings.

The present construction does not define such an assignment. Instead, it defines four context-indexed maps:

$$F^{(a,b)}, \quad F^{(a,b')}, \quad F^{(a',b)}, \quad F^{(a',b')}.$$

These maps are associated with four different measurement Hamiltonians and four different pointer-sector decompositions.

There is therefore no single global map

$$G : \Sigma_{SA}^{(N)} \rightarrow \{\pm 1\}^4$$

assigning all four counterfactual outcomes simultaneously.

This is precisely the structural feature required to avoid the Fine-Bell joint-distribution premise Fine [1982]. The theory assigns definite outcomes to performed measurements, not to all incompatible measurements at once.

## 8.4 CHSH Value in the Pointer-Sector Construction

The CHSH expression may be written as

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b').$$

For the standard coplanar choice of detector settings, quantum mechanics gives

$$|S| = 2\sqrt{2}.$$

In the present construction, each correlation is computed from the corresponding pointer-sector probabilities:

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b).$$

Since

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

we obtain

$$E(a, b) = -a \cdot b.$$

Choosing detector settings such that

$$a \cdot b = a \cdot b' = a' \cdot b = \frac{1}{\sqrt{2}}, \quad a' \cdot b' = -\frac{1}{\sqrt{2}},$$

with the sign convention used in the displayed CHSH expression, gives

$$S = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -2\sqrt{2}.$$

Therefore

$$|S| = 2\sqrt{2}.$$

The sign of  $S$  depends on the labelling convention for the detector settings and outcomes. The physical content is the magnitude

$$|S| = 2\sqrt{2},$$

which exceeds the Bell–CHSH bound

$$|S| \leq 2$$

satisfied by Bell-local hidden-variable models.

There is no contradiction with Bell’s theorem. The four correlations are computed from four context-indexed pointer-sector constructions. They are not derived from a single global joint assignment over

$$(A_a, A_{a'}, B_b, B_{b'}).$$

## 8.5 Relation to the Fine-Bell Analysis

The role of this section is not to reprove the full Fine-Bell theorem. Its purpose is to locate the present system–apparatus construction within the standard taxonomy.

The construction satisfies:

1. definite outcomes for each realised context;
2. operational no-signalling after marginalisation;
3. the singlet correlation  $E(a, b) = -a \cdot b$ ;
4. absence of a global joint assignment for incompatible settings.

It does not satisfy Bell-local outcome factorisation. That is the premise rejected by the context-indexed measurement-sector structure.

The companion Fine-Bell analysis develops this point at the level of contextual outcome maps and joint-distribution constraints. The present note supplies the corresponding system–apparatus construction for the two-qubit singlet measurement.

Thus the result is Bell-consistent: it reproduces the Bell-singlet correlation while explicitly declining the Bell-local factorisation assumption that would make such reproduction impossible.

## 9 Discussion

### 9.1 What Has Been Established

This note has exhibited the standard von Neumann measurement construction for the two-qubit singlet inside the finite-dimensional geometric reconstruction. The result is not a new measurement Hamiltonian, nor an independent derivation of the Born rule. It is a compatibility and closure result: the canonical quantum-pointer measurement model translates directly into the present system–apparatus framework.

The construction establishes the following points.



1. The two-qubit singlet measurement can be represented on a finite-dimensional system–apparatus Hilbert space

$$\mathcal{H}_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathcal{K}_A^{(d_A)} \otimes \mathcal{K}_B^{(d_B)}.$$

2. The corresponding projective state space

$$\mathbb{CP}^{N-1}$$

is compact and carries the standard Kähler structure used in geometric quantum mechanics.

3. The impulsive von Neumann Hamiltonian

$$\hat{H}_{a,b} = \kappa_A(\sigma \cdot a \otimes \mathbb{I}) \otimes \hat{p}_A + \kappa_B(\mathbb{I} \otimes \sigma \cdot b) \otimes \hat{p}_B$$

generates the branch-separated pointer state associated with the measurement context  $(a, b)$ .

4. Pointer-sector projectors

$$\hat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \hat{Q}_A^s \otimes \hat{Q}_B^t$$

define the operational outcome sectors.

5. In the strong-readout limit, the sector weights reduce to the singlet projector probabilities

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

6. Direct summation yields

$$E(a, b) = -a \cdot b.$$

7. Operational no-signalling follows by marginal summation:

$$P_A(\pm 1 | a, b) = P_B(\pm 1 | a, b) = \frac{1}{2}.$$

8. The construction is Bell-consistent because it uses context-indexed outcome sectors rather than Bell-local factorised outcome functions.

The note therefore closes a specific constructive slot: it shows that the non-constructive singlet probabilities obtained upstream are realised by an explicit system–apparatus Hamiltonian and pointer-sector readout model.

## 9.2 Compatibility Rather Than Novel Measurement Theory

The contribution of this note is compatibility, not novelty in the von Neumann measurement model itself. The Hamiltonian, branch-separation calculation, and projector-form probabilities are standard ingredients of quantum measurement theory von Neumann [1932]. What is established here is that these ingredients embed cleanly into the finite-dimensional geometric framework.

This matters because a reconstruction programme must reproduce canonical quantum measurement practice without distortion. If the standard measurement construction could not be represented naturally, that would expose a structural defect in the framework. The fact that it translates directly, with no Fubini–Study Voronoi-volume construction and no additional probability rule, is evidence that the finite-dimensional architecture can absorb standard measurement theory rather than replacing it with an ad hoc alternative.

The calculation also demonstrates ease of use. Once the system–apparatus Hilbert space and pointer projectors are specified, the computation proceeds exactly as in the textbook treatment. The geometric interpretation changes the conceptual placement of the calculation, not the operational calculation itself.

The result should therefore be read as an architectural closure result. Technical Note 2 supplied the non-constructive singlet probability calculation. The present note supplies the corresponding system–apparatus realisation. Together, they show that the Bell-singlet sector is covered both at the probability-functional level and at the measurement-model level.

### 9.3 Relation to Bohmian Branch Selection

The construction has a clear structural parallel with Bohmian mechanics applied to a Bell-singlet measurement Bohm [1952b], Dürr et al. [1992]. Both frameworks use deterministic ontology, branch selection, and standard operator-form Born probabilities for the measurement statistics.

In Bohmian mechanics, the actual configuration of the system and apparatus lies in one branch of the post-measurement wavefunction. The other branches remain part of the wavefunction but do not contain the actual configuration. In the present framework, the analogous statement is that the ontic state lies in one realised outcome sector associated with the measurement context. The operator-form weights determine the preparation-relative sector weights, while the programme-level ontology supplies single-outcome definiteness.

The distinction is that the present framework expresses the system–apparatus dynamics through symplectic Hamiltonian flow on projective state space in the geometric quantum mechanics formulation, with the wider ontic interpretation organised through the projection

$$\pi_{SA}^{(N)} : \Sigma_{SA}^{(N)} \rightarrow \mathbb{CP}^{N-1}.$$

Whether this difference amounts to a deeper theoretical distinction belongs to the broader programme rather than to the present technical note.

For the purpose of this note, the important point is narrower: the standard branch-selection structure of a quantum-pointer measurement is compatible with the finite-dimensional geometric architecture.

### 9.4 Limitations and Non-Claims

Several limitations should be stated explicitly.

First, the construction does not derive the Born rule. The probability rule is inherited from the upstream finite-dimensional Born-weight layer. The present note shows how those probabilities arise as pointer-sector weights in a standard system–apparatus model.

Second, the construction does not derive the singlet probability kernel from first principles. The expression

$$P_{st}(a, b) = \frac{1 - st \, a \cdot b}{4}$$

is obtained from the usual singlet projector calculation, as already used in Technical Note 2.

Third, the construction is not Bell-local. It does not define factorised outcome functions

$$A(a, \lambda), \quad B(b, \lambda).$$

Instead, it defines outcome sectors for each realised measurement context  $(a, b)$ . This is necessary for compatibility with Bell-type constraints.

Fourth, the pointer sectors are finite-dimensional truncations. The continuous pointer model is treated only as a controlled comparison limit. Exact statements are made at fixed finite truncation.

Fifth, the note does not provide a general classification of all physically realisable measurement partitions. It supplies one explicit canonical construction: the von Neumann quantum-pointer measurement model for the two-qubit singlet.

Sixth, the note does not by itself solve the full measurement problem. It shows how standard branch separation and pointer readout are represented in the finite-dimensional geometric architecture. The deeper ontological account of single-outcome branch selection belongs to the wider framework.

### 9.5 Role Within the Finite-Dimensional Reconstruction

Within the finite-dimensional reconstruction, the present note has a specific role. It sits downstream of the measure-and-weight layer and upstream of later formalisation of measurement-control structures.

The dependency chain is:

1. the repeated-trial and typicality layer supplies the bridge from invariant weights to long-run frequencies;
2. the measure and Born-weight layer supplies the finite-dimensional probability rule;
3. Technical Note 2 applies that rule to the singlet and obtains the Bell correlation non-constructively;

4. the present note embeds the standard system–apparatus measurement construction that realises those probabilities dynamically;
5. later formalisation work, assigned to LF3, can formalise the control-Hamiltonian and outcome-sector structure.

The note therefore performs constructive closure for the singlet specialisation. It does not replace the probability layer, and it does not extend the theory to field-theoretic or continuum regimes. Its contribution is to show that the canonical finite-dimensional measurement construction fits cleanly into the programme architecture.

In this sense, the note functions as a compatibility test. The framework must be able to reproduce the standard measurement model without distortion. The construction above shows that it can, at least for the two-qubit singlet in the finite-dimensional quantum-pointer setting.

## 10 Conclusion

This note has exhibited the standard von Neumann quantum-pointer measurement construction for the two-qubit singlet inside the finite-dimensional geometric reconstruction. The construction begins with a finite system–apparatus Hilbert space,

$$\mathcal{H}_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathcal{K}_A^{(d_A)} \otimes \mathcal{K}_B^{(d_B)},$$

and uses the corresponding compact projective state space

$$\mathbb{CP}^{N-1}$$

as the geometric arena for Schrödinger evolution, equivalently Hamiltonian flow in the Ashtekar–Schilling formulation.

For detector settings

$$a, b \in S^2,$$

the impulsive measurement Hamiltonian is

$$\hat{H}_{a,b} = \kappa_A(\sigma \cdot a \otimes \mathbb{I}) \otimes \hat{p}_A + \kappa_B(\mathbb{I} \otimes \sigma \cdot b) \otimes \hat{p}_B.$$

Because the two readout couplings commute, the measurement unitary factorises into independent pointer translations conditioned on the eigenvalues of

$$\sigma \cdot a \quad \text{and} \quad \sigma \cdot b.$$

Applied to the initial singlet-pointer state,

$$|\Psi_0\rangle = |\psi^-\rangle \otimes |\phi_A^0\rangle \otimes |\phi_B^0\rangle,$$

this interaction produces the branch-separated state

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a,b) |s_a, t_b\rangle \otimes |\phi_A^s(T)\rangle \otimes |\phi_B^t(T)\rangle.$$

The outcome sectors are defined by pointer projectors,

$$\hat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \hat{Q}_A^s \otimes \hat{Q}_B^t,$$

rather than by Fubini–Study Voronoi regions. Their weights are operator-form branch weights:

$$P_{st}(a,b;T) = \langle \Psi_T | \hat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

Under the branch-separation assumptions, these weights satisfy

$$P_{st}(a,b;T) = |c_{st}(a,b)|^2 + O(\epsilon_A + \epsilon_B).$$

In the strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

the weights reduce to the standard singlet projector probabilities:

$$P_{st}(a,b) = \frac{1 - st a \cdot b}{4}.$$

It follows immediately that

$$E(a,b) = \sum_{s,t \in \{\pm 1\}} st P_{st}(a,b) = -a \cdot b.$$

The local marginals satisfy

$$P_A(\pm 1 | a, b) = P_B(\pm 1 | a, b) = \frac{1}{2},$$

so operational no-signalling follows by direct marginal summation.

The construction is deterministic per realised measurement context, but it is not Bell-local. It uses context-indexed outcome sectors

$$F^{(a,b)} : \Sigma_{SA}^{(N)} \rightarrow \{\pm 1\} \times \{\pm 1\},$$

not factorised hidden-variable outcome functions of the form

$$A(a, \lambda), \quad B(b, \lambda).$$

Thus the construction is compatible with Bell-type constraints while reproducing the Bell-singlet correlation.

The contribution of the note is therefore precise. It does not introduce a novel measurement Hamiltonian, derive the Born rule, or solve the general measurement problem. It shows that the canonical von Neumann measurement construction embeds cleanly into the finite-dimensional geometric architecture and supplies the constructive system–apparatus realisation of the singlet probabilities obtained upstream. This establishes compatibility and architectural closure for the two-qubit singlet sector and provides a formalisation-ready target for the later LF3 measurement-control layer.

## A Finite-Dimensional Pointer Truncation

### A.1 Truncated Pointer Hilbert Spaces

The ideal von Neumann pointer is usually represented by a continuous position degree of freedom with Hilbert space

$$L^2(\mathbb{R}).$$

In that setting, the pointer momentum operator generates translations of the pointer wavepacket, and a measurement interaction shifts the packet into macroscopically distinguishable position regions.

The present note uses a finite-dimensional truncation of this construction. For each wing, we take finite-dimensional pointer spaces

$$\mathcal{K}_A^{(d_A)}, \quad \mathcal{K}_B^{(d_B)},$$

with

$$d_A, d_B < \infty.$$

The total system–apparatus Hilbert space is therefore finite-dimensional:

$$\mathcal{H}_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathcal{K}_A^{(d_A)} \otimes \mathcal{K}_B^{(d_B)}, \quad N = 4d_A d_B.$$

The corresponding pure-state space is the compact projective manifold

$$\mathbb{CP}^{N-1}.$$

This is the finite-dimensional setting in which all exact statements in the present note are made.

The finite pointer spaces may be understood as detector windows with a large but finite number of resolved position modes. In this approximation, the continuous pointer packet is replaced by a vector supported on a finite position grid or finite wavepacket basis. The two readout outcomes correspond to two disjoint subspaces:

$$\mathcal{K}_A^{(+)}, \quad \mathcal{K}_A^{(-)}$$

for the  $A$ -pointer, and

$$\mathcal{K}_B^{(+)}, \quad \mathcal{K}_B^{(-)}$$

for the  $B$ -pointer.

The associated projectors are

$$\hat{Q}_A^+, \hat{Q}_A^-, \quad \hat{Q}_B^+, \hat{Q}_B^-.$$

They satisfy

$$\hat{Q}_A^+ + \hat{Q}_A^- = \mathbb{I}_A^P, \quad \hat{Q}_A^+ \hat{Q}_A^- = 0,$$

and

$$\hat{Q}_B^+ + \hat{Q}_B^- = \mathbb{I}_B^P, \quad \hat{Q}_B^+ \hat{Q}_B^- = 0.$$

## A.2 Approximation to Continuous Position Pointers

The finite-dimensional truncation is required only to reproduce the operational features used in the measurement calculation. These are:

1. an initially localised neutral pointer state;
2. a self-adjoint generator of approximate pointer translation;
3. two disjoint readout subspaces corresponding to outcomes  $+1$  and  $-1$ ;
4. controllable leakage between the intended and unintended readout sectors.

Let

$$|\phi_A^0\rangle \in \mathcal{K}_A^{(d_A)}$$

be the initial  $A$ -pointer state. The measurement interaction translates this state into

$$|\phi_A^s(T)\rangle = \exp\left(-\frac{i}{\hbar}s\kappa_A T\hat{p}_A\right)|\phi_A^0\rangle, \quad s \in \{\pm 1\}.$$

The required approximation condition is

$$\langle\phi_A^s(T)|\hat{Q}_A^s|\phi_A^s(T)\rangle \geq 1 - \epsilon_A,$$

and

$$\langle\phi_A^s(T)|\hat{Q}_A^{-s}|\phi_A^s(T)\rangle \leq \epsilon_A.$$

Similarly, for the  $B$ -pointer,

$$|\phi_B^t(T)\rangle = \exp\left(-\frac{i}{\hbar}t\kappa_B T\hat{p}_B\right)|\phi_B^0\rangle, \quad t \in \{\pm 1\},$$

with

$$\langle\phi_B^t(T)|\hat{Q}_B^t|\phi_B^t(T)\rangle \geq 1 - \epsilon_B,$$

and

$$\langle\phi_B^t(T)|\hat{Q}_B^{-t}|\phi_B^t(T)\rangle \leq \epsilon_B.$$

The parameters

$$\epsilon_A, \epsilon_B$$

measure finite truncation and finite readout imperfections. In the ideal continuous pointer model, these errors can be made arbitrarily small by increasing pointer separation relative to packet width. In the finite-dimensional setting, the same role is played by increasing the truncation size and choosing the detector windows so that the translated packets remain well separated.

For a Gaussian pointer packet of width  $\Delta q_A$ , translated by a displacement

$$D_A = \kappa_A T,$$

the leakage into the wrong readout sector decreases as the separation ratio

$$D_A/\Delta q_A$$

increases. In a finite truncation, the detector window must be large enough to contain the translated packets without boundary effects, so one requires a window size  $L_A$  satisfying

$$L_A \gg D_A + \Delta q_A.$$

Equivalently, the pointer dimension  $d_A$  must be large enough to resolve both packets and their separation. The same considerations apply to  $B$ . Thus, for fixed packet width and readout threshold, the leakage parameters

$$\epsilon_A, \epsilon_B$$

can be made arbitrarily small by increasing the pointer displacement and choosing sufficiently large finite pointer truncations.

### A.3 Compactness of the Joint Projective Manifold

The use of finite pointer truncations ensures that the joint projective state space is compact:

$$\mathbb{CP}^{N-1}, \quad N = 4d_A d_B.$$

This is important for consistency with the finite-dimensional reconstruction, where the exact geometric statements are made on compact projective spaces.

The continuous model

$$\mathbb{C}^4 \otimes L^2(\mathbb{R}) \otimes L^2(\mathbb{R})$$

is not the exact setting of the present note. It serves only as the physical motivation for the finite-dimensional pointer approximation.

The finite-dimensional formulation should therefore be read as follows:

1. exact claims are made at fixed finite  $d_A, d_B$ ;
2. finite pointer imperfections are controlled by  $\epsilon_A, \epsilon_B$ ;
3. the continuous pointer model is recovered only as a comparison limit;
4. compactness is preserved throughout the exact finite-dimensional construction.

Thus the note remains within compact finite-dimensional projective geometry while retaining the operational content of the standard von Neumann pointer measurement.

## B Finite-Separation Error Bounds

### B.1 Pointer Leakage Parameters

The ideal readout limit assumes that the translated pointer packets lie entirely inside their intended readout sectors. At finite pointer separation, there may be a small leakage into the wrong sector.

For the  $A$ -pointer, define

$$\epsilon_A = \max_{s \in \{\pm 1\}} \langle \phi_A^s(T) | \widehat{Q}_A^{-s} | \phi_A^s(T) \rangle.$$

Equivalently,

$$\langle \phi_A^s(T) | \widehat{Q}_A^s | \phi_A^s(T) \rangle \geq 1 - \epsilon_A.$$

For the  $B$ -pointer, define

$$\epsilon_B = \max_{t \in \{\pm 1\}} \langle \phi_B^t(T) | \widehat{Q}_B^{-t} | \phi_B^t(T) \rangle,$$

so that

$$\langle \phi_B^t(T) | \widehat{Q}_B^t | \phi_B^t(T) \rangle \geq 1 - \epsilon_B.$$

These leakage parameters quantify the finite separation of pointer branches. They include the effects of finite truncation, imperfect pointer localisation, nonzero packet tails, and any residual spreading neglected in the impulsive approximation.

The ideal strong-readout limit is

$$\epsilon_A, \epsilon_B \rightarrow 0.$$

### B.2 Error Bound for Branch Weights

The post-measurement state is

$$|\Psi_T\rangle = \sum_{u,v \in \{\pm 1\}} c_{uv}(a, b) |u_a, v_b\rangle \otimes |\phi_A^u(T)\rangle \otimes |\phi_B^v(T)\rangle.$$

The pointer-sector probability for outcome  $(s, t)$  is

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle,$$

with

$$\widehat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \widehat{Q}_A^s \otimes \widehat{Q}_B^t.$$

Using orthogonality of the system eigenvectors, this becomes

$$P_{st}(a, b; T) = \sum_{u,v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \alpha_A(s|u) \alpha_B(t|v),$$

where

$$\alpha_A(s|u) = \langle \phi_A^u(T) | \widehat{Q}_A^s | \phi_A^u(T) \rangle,$$

and

$$\alpha_B(t|v) = \langle \phi_B^v(T) | \widehat{Q}_B^t | \phi_B^v(T) \rangle.$$

By definition of the leakage parameters,

$$\alpha_A(u|u) \geq 1 - \epsilon_A, \quad \alpha_A(-u|u) \leq \epsilon_A,$$

and

$$\alpha_B(v|v) \geq 1 - \epsilon_B, \quad \alpha_B(-v|v) \leq \epsilon_B.$$

The ideal branch weight is

$$P_{st}^{\text{ideal}}(a, b) = |c_{st}(a, b)|^2.$$

We now bound the deviation

$$\Delta_{st}(a, b; T) = P_{st}(a, b; T) - P_{st}^{\text{ideal}}(a, b).$$

The intended branch  $(s, t)$  contributes

$$|c_{st}|^2 \alpha_A(s|s) \alpha_B(t|t).$$

Its deviation from the ideal contribution  $|c_{st}|^2$  is

$$|c_{st}|^2 [1 - \alpha_A(s|s) \alpha_B(t|t)].$$

Since

$$\alpha_A(s|s) \geq 1 - \epsilon_A, \quad \alpha_B(t|t) \geq 1 - \epsilon_B,$$

we have

$$\alpha_A(s|s) \alpha_B(t|t) \geq (1 - \epsilon_A)(1 - \epsilon_B).$$

Therefore

$$1 - \alpha_A(s|s) \alpha_B(t|t) \leq \epsilon_A + \epsilon_B - \epsilon_A \epsilon_B.$$

The intended-branch deficit is consequently bounded by

$$|c_{st}|^2 (\epsilon_A + \epsilon_B - \epsilon_A \epsilon_B).$$

It remains to bound the wrong-branch contributions. There are three such terms:

$$(s, -t), \quad (-s, t), \quad (-s, -t).$$

For the branch  $(s, -t)$ , the  $A$ -pointer factor is correctly aligned but the  $B$ -pointer factor is in the wrong readout sector. Hence

$$|c_{s,-t}|^2 \alpha_A(s|s) \alpha_B(t|-t) \leq |c_{s,-t}|^2 \epsilon_B.$$

For the branch  $(-s, t)$ , the  $B$ -pointer factor is correctly aligned but the  $A$ -pointer factor is in the wrong readout sector. Hence

$$|c_{-s,t}|^2 \alpha_A(s|-s) \alpha_B(t|t) \leq |c_{-s,t}|^2 \epsilon_A.$$

For the branch  $(-s, -t)$ , both pointer factors are in the wrong readout sectors. Hence

$$|c_{-s,-t}|^2 \alpha_A(s|-s) \alpha_B(t|-t) \leq |c_{-s,-t}|^2 \epsilon_A \epsilon_B.$$

Combining these estimates gives

$$\begin{aligned} |P_{st}(a, b; T) - |c_{st}(a, b)|^2| &\leq |c_{st}|^2 (\epsilon_A + \epsilon_B - \epsilon_A \epsilon_B) \\ &\quad + |c_{s,-t}|^2 \epsilon_B + |c_{-s,t}|^2 \epsilon_A + |c_{-s,-t}|^2 \epsilon_A \epsilon_B. \end{aligned}$$

Using

$$\sum_{u,v \in \{\pm 1\}} |c_{uv}|^2 = 1,$$

this implies the simpler uniform estimate

$$|P_{st}(a, b; T) - |c_{st}(a, b)|^2| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

In particular,

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + O(\epsilon_A + \epsilon_B).$$

In the strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

we obtain

$$P_{st}(a, b; T) \rightarrow |c_{st}(a, b)|^2.$$

### B.3 Error Bound for the Correlator

The finite-separation correlation is

$$E_T(a, b) = \sum_{s,t \in \{\pm 1\}} st P_{st}(a, b; T).$$

Write

$$P_{st}(a, b; T) = P_{st}^{\text{ideal}}(a, b) + \Delta_{st}(a, b; T),$$

where

$$\Delta_{st}(a, b; T) = O(\epsilon_A + \epsilon_B).$$

Then

$$E_T(a, b) = \sum_{s,t} st P_{st}^{\text{ideal}}(a, b) + \sum_{s,t} st \Delta_{st}(a, b; T).$$

The first term is the ideal singlet correlation:

$$\sum_{s,t} st P_{st}^{\text{ideal}}(a, b) = -a \cdot b.$$

The second term is bounded by

$$\left| \sum_{s,t} st \Delta_{st}(a, b; T) \right| \leq \sum_{s,t} |\Delta_{st}(a, b; T)|.$$

Since each

$$\Delta_{st}(a, b; T) = O(\epsilon_A + \epsilon_B),$$

we obtain

$$E_T(a, b) = -a \cdot b + O(\epsilon_A + \epsilon_B).$$

Thus the Bell-singlet correlation is recovered exactly in the strong-readout limit and approximately at finite pointer separation.

### B.4 Error Bound for Marginals

The finite-separation marginal at  $A$  is

$$P_A(s | a, b; T) = \sum_{t \in \{\pm 1\}} P_{st}(a, b; T).$$

Using

$$P_{st}(a, b; T) = P_{st}^{\text{ideal}}(a, b) + \Delta_{st}(a, b; T),$$

we obtain

$$P_A(s | a, b; T) = \sum_t P_{st}^{\text{ideal}}(a, b) + \sum_t \Delta_{st}(a, b; T).$$



The ideal marginal is

$$\sum_t P_{st}^{\text{ideal}}(a, b) = \frac{1}{2}.$$

Therefore,

$$P_A(s | a, b; T) = \frac{1}{2} + O(\epsilon_A + \epsilon_B).$$

Similarly,

$$P_B(t | a, b; T) = \frac{1}{2} + O(\epsilon_A + \epsilon_B).$$

Thus operational no-signalling is exact in the strong-readout limit and stable under finite branch-separation errors.

### B.5 Interpretation of the Error Terms

The leakage parameters

$$\epsilon_A, \quad \epsilon_B$$

do not modify the underlying singlet probability law. They quantify imperfections in the pointer readout stage.

Physically, these errors may arise from:

1. insufficient pointer displacement during the readout interval;
2. overlap between translated pointer packets;
3. finite truncation of the pointer Hilbert space;
4. residual free spreading neglected in the impulsive idealisation;
5. imperfect alignment between pointer packets and readout projectors.

In the idealised projective measurement limit, these effects vanish:

$$\epsilon_A, \epsilon_B \rightarrow 0.$$

The exact probabilities are then

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

the exact correlation is

$$E(a, b) = -a \cdot b,$$

and the exact marginals are

$$P_A(\pm 1 | a, b) = P_B(\pm 1 | a, b) = \frac{1}{2}.$$

The finite-separation analysis therefore shows that the construction is not dependent on mathematically perfect pointer separation. The ideal result is approached continuously as branch leakage is reduced.

## C Relation to the CSD Dependency Stack

### C.1 Relation to Technical Note 2

Technical Note 2 supplies the non-constructive singlet probability calculation used in the present note. Its role is to show that the two-qubit singlet lies within the finite-dimensional Born-weight layer and that the joint spin probabilities are

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

The present note does not rederive this probability rule from first principles. Instead, it takes the TN2 probability kernel as the upstream finite-dimensional result and shows how it is realised by a standard quantum-pointer measurement Hamiltonian.

The division of labour is therefore:

1. TN2 evaluates the singlet probability kernel at the operator-probability level.
2. The present note constructs the corresponding system–apparatus measurement model.
3. The pointer-sector calculation shows that the TN2 probabilities arise as branch-sector weights after von Neumann readout.

This makes the present note the constructive companion to TN2. TN2 establishes the result at the probability-functional level; this note embeds the same result in an explicit measurement interaction.

There is one important qualification. TN2 Appendix B.8 records H5 as an operational consistency condition and states that dynamical verification is deferred to TN3. The present note discharges only a weak version of that promise. Section 5.6 verifies that, for the von Neumann pointer construction used here, a shared macroscopic pointer channel represented by the same tensor-factor projector has the same associated pullback sector across compatible contexts that differ only in the remote completion. This is sufficient for the shared-channel consistency used in the present singlet construction. It is not a full proof of Gleason-style noncontextuality across arbitrary orthogonal completions, nor a classification theorem for all dynamically realisable apparatus partitions.

## C.2 Relation to Paper B

Paper B supplies the finite-dimensional Born-weight layer. In particular, it explains why the admissible finite-dimensional probability assignment has the operator form

$$P(i|\rho) = \text{Tr}(\rho\Pi_i)$$

for projective outcomes, under the stated symmetry and operational consistency assumptions.

The present note uses that layer in the specific case of the two-qubit singlet. The probability assigned to the pointer-sector projector

$$\widehat{M}_{st}^{(a,b)}$$

is

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

This is not an additional probability postulate. It is the operator-form probability rule inherited from the finite-dimensional Born-weight layer. The novelty of the present note is not the probability rule itself, but the explicit system–apparatus realisation showing how the singlet projector probabilities appear as pointer-sector branch weights after measurement interaction.

Thus Paper B supplies the probability rule; TN2 applies it to the singlet; the present note supplies the measurement Hamiltonian that realises the resulting branch-sector structure.

## C.3 Relation to TN4

TN4 develops the broader finite-dimensional operational dictionary, including mixed states, POVMs, reduced states, and subsystem representations. The present note uses only the projective, pure-state part of that dictionary.

The relevant ingredients are:

1. a finite-dimensional pure state

$$|\Psi_T\rangle \in \mathcal{H}_{SA}^{(N)};$$

2. projective pointer readout sectors

$$\widehat{M}_{st}^{(a,b)} = \mathbb{I}_{AB} \otimes \widehat{Q}_A^s \otimes \widehat{Q}_B^t;$$

3. operator-form branch weights

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

No general POVM machinery is required for the idealised strong-readout calculation. However, TN4 becomes relevant for more realistic detector models, where pointer readout may be noisy, coarse-grained, inefficient, or represented by effects rather than sharp projectors.

In this sense, the present note is narrower than TN4. It gives a clean projective measurement realisation of the singlet sector, while TN4 supplies the more general operational machinery needed for non-ideal measurements and subsystem descriptions.

#### C.4 Relation to c1

The companion Fine-Bell analysis c1 supplies the broader no-go-theorem diagnosis. The present note uses the same structural distinction:

definite outcome per realised context  
does not imply  
a single global assignment across all incompatible contexts.

For each measurement context

$$(a, b),$$

the present note defines context-indexed outcome sectors

$$F^{(a,b)} : \Sigma_{SA}^{(N)} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

It does not define a global map

$$G : \Sigma_{SA}^{(N)} \rightarrow \{\pm 1\}^4$$

assigning simultaneous outcomes for

$$(a, b), \quad (a, b'), \quad (a', b), \quad (a', b').$$

This is the structural reason the construction is compatible with Bell-type constraints. It reproduces the singlet correlations, including the CHSH value

$$|S| = 2\sqrt{2},$$

because it does not assume Bell-local outcome factorisation or a global Fine-style joint distribution.

The present note therefore provides the system–apparatus realisation of the contextual structure analysed abstractly in c1.

#### C.5 Relation to LF3

The present note is not a Lean formalisation paper. Its formalisation target is deferred to LF3.

Nevertheless, the construction has been written so that LF3 has a precise theorem schema to formalise. The key ingredients are finite-dimensional:

1. the system Hilbert space

$$\mathbb{C}^2 \otimes \mathbb{C}^2;$$

2. finite-dimensional pointer spaces

$$\mathcal{K}_A^{(d_A)}, \quad \mathcal{K}_B^{(d_B)};$$

3. the joint finite-dimensional space

$$\mathcal{H}_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathcal{K}_A^{(d_A)} \otimes \mathcal{K}_B^{(d_B)};$$

4. the setting-dependent Hamiltonian

$$\hat{H}_{a,b} = \kappa_A(\sigma \cdot a \otimes \mathbb{I}) \otimes \hat{p}_A + \kappa_B(\mathbb{I} \otimes \sigma \cdot b) \otimes \hat{p}_B;$$

5. the factorised unitary evolution generated by commuting readout terms;

6. the branch-separated final state;

7. pointer-sector projectors

$$\hat{M}_{st}^{(a,b)};$$

8. leakage parameters

$$\epsilon_A, \epsilon_B;$$

9. the branch-weight estimate

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + O(\epsilon_A + \epsilon_B);$$

10. the singlet specialisation

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4};$$

11. the resulting correlation and marginals:

$$E(a, b) = -a \cdot b, \quad P_A(\pm 1 | a, b) = P_B(\pm 1 | a, b) = \frac{1}{2}.$$

LF3 should therefore formalise the measurement-control and outcome-sector layer, not the upstream Born-weight derivation. The upstream layers belong to LF1 and LF2. The present note prepares the LF3 target by isolating the finite-dimensional objects, assumptions, and calculations needed for that later formalisation.

## C.6 Dependency Summary

The logical role of the present note can be summarised as follows:

$$\text{Paper B} \longrightarrow \text{TN2} \longrightarrow \text{TN3} \longrightarrow \text{LF3}.$$

More explicitly:

1. Paper B supplies the finite-dimensional Born-weight rule.
2. TN2 applies that rule to the two-qubit singlet and obtains

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

3. The present note realises those probabilities through a von Neumann quantum-pointer measurement Hamiltonian.
4. LF3 will formalise the measurement-control and outcome-sector structure.

The present note therefore closes the constructive singlet layer without claiming to replace the upstream probability derivation or the downstream formalisation.

## References

- Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13(11):113036, November 2011. ISSN 1367-2630. doi:10.1088/1367-2630/13/11/113036. URL <https://doi.org/10.1088/1367-2630/13/11/113036>.
- Vladimir I. Arnold. *Mathematical Methods of Classical Mechanics*. Springer, New York, 2 edition, 1989. ISBN 9780387968902. doi:10.1007/978-1-4757-2063-1. URL <https://doi.org/10.1007/978-1-4757-2063-1>.
- Abhay Ashtekar and Troy A. Schilling. Geometrical formulation of quantum mechanics. In Alex Harvey, editor, *On Einstein's Path*, pages 23–65. Springer New York, 1999. ISBN 9781461214229. doi:10.1007/978-1-4612-1422-9\_3. URL [https://doi.org/10.1007/978-1-4612-1422-9\\_3](https://doi.org/10.1007/978-1-4612-1422-9_3).
- John S. Bell. On the einstein podolsky rosen paradox. *Physics Physique Fizika*, 1(3):195–200, November 1964. ISSN 0554-128X. doi:10.1103/PhysicsPhysiqueFizika.1.195. URL <https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195>.
- Zayn Blore. Fine-bell consistency of contextual outcome maps in constraint-surface dynamics, 2026a. Preprint.
- Zayn Blore. Fixing the measure: Deriving born-form weights from symmetry in deterministic geometry, 2026b. Preprint.
- Zayn Blore. Tn2: The two-qubit singlet correlation as a symmetry-and-operations specialisation of deterministic volume geometry, 2026c. Preprint.
- Zayn Blore. Tn4: Mixed states, povms, and subsystem reduction in constraint-surface dynamics, 2026d. Preprint.
- David Bohm. A suggested interpretation of the quantum theory in terms of “hidden” variables. i. *Physical Review*, 85(2):166–179, January 1952a. ISSN 0031-899X. doi:10.1103/PhysRev.85.166. URL <https://doi.org/10.1103/PhysRev.85.166>.
- David Bohm. A suggested interpretation of the quantum theory in terms of “hidden” variables. ii. *Physical Review*, 85(2):180–193, January 1952b. ISSN 0031-899X. doi:10.1103/PhysRev.85.180. URL <https://doi.org/10.1103/PhysRev.85.180>.

- Max Born. Quantenmechanik der stoßvorgänge. *Zeitschrift für Physik*, 38(11–12):803–827, November 1926. ISSN 1434-601X. doi:10.1007/BF01397184. URL <https://doi.org/10.1007/BF01397184>.
- Paul Busch. Quantum states and generalized observables: A simple proof of gleason’s theorem. *Physical Review Letters*, 91(12):120403, September 2003. ISSN 0031-9007. doi:10.1103/PhysRevLett.91.120403. URL <https://doi.org/10.1103/PhysRevLett.91.120403>.
- John F. Clauser, Michael A. Horne, Abner Shimony, and Richard A. Holt. Proposed experiment to test local hidden-variable theories. *Physical Review Letters*, 23(15):880–884, October 1969. ISSN 0031-9007. doi:10.1103/PhysRevLett.23.880. URL <https://doi.org/10.1103/PhysRevLett.23.880>.
- Detlef Dürr, Sheldon Goldstein, and Nino Zanghì. Quantum equilibrium and the origin of absolute uncertainty. *Journal of Statistical Physics*, 67(5–6):843–907, June 1992. ISSN 0022-4715. doi:10.1007/BF01049004. URL <https://doi.org/10.1007/BF01049004>.
- Detlef Dürr, Sheldon Goldstein, and Nino Zanghì. *Quantum Physics Without Quantum Philosophy*. Springer, Berlin, 2013. ISBN 9783642306890. doi:10.1007/978-3-642-30690-7. URL <https://doi.org/10.1007/978-3-642-30690-7>.
- Arthur Fine. Hidden variables, joint probability, and the bell inequalities. *Physical Review Letters*, 48(5):291–295, February 1982. ISSN 0031-9007. doi:10.1103/PhysRevLett.48.291. URL <https://doi.org/10.1103/PhysRevLett.48.291>.
- Andrew M. Gleason. Measures on the closed subspaces of a hilbert space. *Journal of Mathematics and Mechanics*, 6(4):885–893, 1957. ISSN 0003-486X. doi:10.2307/1969989. URL <https://doi.org/10.2307/1969989>.
- Brian C. Hall. *Quantum Theory for Mathematicians*. Springer, New York, 2013. ISBN 9781461471158. doi:10.1007/978-1-4614-7116-5. URL <https://doi.org/10.1007/978-1-4614-7116-5>.
- Thomas W. B. Kibble. Geometrization of quantum mechanics. *Communications in Mathematical Physics*, 65(2):189–201, 1979. ISSN 0010-3616. doi:10.1007/BF01225149. URL <https://doi.org/10.1007/BF01225149>.
- Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967. ISSN 0022-2518. doi:10.1512/iumj.1968.17.17004. URL <https://doi.org/10.1512/iumj.1968.17.17004>.
- Matthew F. Pusey, Jonathan Barrett, and Terry Rudolph. On the reality of the quantum state. *Nature Physics*, 8(6):475–478, May 2012. ISSN 1745-2473. doi:10.1038/nphys2309. URL <https://doi.org/10.1038/nphys2309>.
- Maximilian Schlosshauer. *Decoherence and the Quantum-to-Classical Transition*. Springer, Berlin, 2007. ISBN 9783540357735. doi:10.1007/978-3-540-35775-9. URL <https://doi.org/10.1007/978-3-540-35775-9>.
- Robert W. Spekkens. Evidence for the epistemic view of quantum states: A toy theory. *Physical Review A*, 75(3):032110, March 2007. ISSN 1050-2947. doi:10.1103/PhysRevA.75.032110. URL <https://doi.org/10.1103/PhysRevA.75.032110>.
- John von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, Berlin, 1932.
- John von Neumann. *Mathematical Foundations of Quantum Mechanics*. Princeton University Press, Princeton, 1955.
- William K. Wootters. Statistical distance and hilbert space. *Physical Review D*, 23(2):357–362, January 1981. ISSN 0556-2821. doi:10.1103/PhysRevD.23.357. URL <https://doi.org/10.1103/PhysRevD.23.357>.
- Wojciech H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3):715–775, May 2003. ISSN 0034-6861. doi:10.1103/RevModPhys.75.715. URL <https://doi.org/10.1103/RevModPhys.75.715>.